

HRITHIK CHANDRA

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PROFESSIONAL SUMMARY

A curious and driven final year BSc (Hons) Computer Science (AI specialization) student specialising in machine learning, data science, and applied AI. I have hands-on experience building ML pipelines, synthetic data simulators, and graph-based models for gene expression prediction, as well as NLP sentiment systems and evaluation tooling from internship and coursework projects.

I combine strong technical skills in Python, SQL, and modern ML libraries with leadership and communication experience as a Student Representative, HertSquad Activator, and Student Guide, being meticulous, organized, highly adaptable, thriving in collaborative, high-performance environments.

My interests span across industries such as - technology, finance, market research, bioinformatics, and automotive, underpinned by a need to build products that endure, adapt, and elevate performance. I seek to craft a career that fuses autonomy with impact, aiming to contribute to the future of intelligent systems and impactful research.

Currently seeking relevant internships, entry level roles with the goal of mastering the powerful AI ML and Data Science tools, full-stack system ownership, deploying ML workflows, and creating sustainable solutions.

EDUCATION

Bachelor of Science in Computer Science (AI Specialization), University of Hertfordshire
High School Diploma (CBSE + AISSCE), New Horizon Scholars School, Thane, India

Sept 2023 – Sept 2026
Mar 2016 – Mar 2022

EXPERIENCE

Student Guide –University of Hertfordshire, Sep 2025 – Aug 2026

- Guided new students around campus, signposting them to relevant services and providing clear information on what to expect during their transition to university life
- Helped create a welcoming and inclusive environment for incoming students, offering reassurance and practical support during their first days on campus.
- Organised and coordinated social events including movie nights, karaoke evenings, and day trips to destinations such as London, Oxford, Cambridge, and Brighton, fostering community and student wellbeing.

Work Experience at Ipsos UK, Aug 2025 – Sept 2025

- Built a reproducible evaluation pipeline for synthetic tabular data using Python (pandas, numpy, scikit-learn), applying OOP design principles, modular code structure, and data hygiene practices to produce auditable, comparable outputs.
- Developed preprocessing and analysis scripts in VS Code and Jupyter Notebook, iterating rapidly on experimental designs to compare model and hyperparameter performance.
- Gained hands-on insight into production-grade ML workflows and NLP pipelines; understood trade-offs relevant to scalable, real-world data systems.
- Communicated findings clearly to both technical and non-technical stakeholders, using Slack and structured documentation, mirroring day-to-day SDE team collaboration.

HertSquad Activator at University of Hertfordshire, Sept 2024 – July 2025

- Fostered a welcoming, inclusive environments and coordinate student engagement in sports, wellness, and social programs across campus.
- Built collaborative relationships with students and university staff; identified barriers to participation and implemented targeted outreach strategies.
- This demonstrates my adaptability in fast-paced environments, while balancing multiple priorities and deadlines.
- Responded promptly to queries and requests from students and staff via email and messaging tools, maintaining clear and professional communication across multiple ongoing workstreams.

Hertfordshire Student's Union – Student Representative, Oct 2023 – Aug 2024

- Acted as liaison between students, academic staff, and the Students' Union; identified issues and supported resolution through cross-functional collaboration.
 - Represented student concerns in formal meetings, documented outcomes, and coordinated follow-up actions across stakeholders.
 - Worked towards improving student-staff engagement improvements by working with fellow representatives and university teams
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PROJECTS

Predicting Gene Expression using ML — Final Year Project (In Progress) - Level 6

- Investigating whether encoding Gene Regulatory Network (GRN) topology as a structural prior improves machine learning models for gene expression prediction over pure tabular approaches.
- Developed a custom continuous-time in-silico simulator using Python and ordinary differential equations (ODEs) to generate synthetic transcriptomic data paired with ground-truth causal GRN graphs.
- Engineered topological network features (node degree, centrality) using NetworkX and developed a reproducible data processing pipeline using Pandas and Scikit-learn.
- Implemented, trained, and compared tabular baselines (Ridge Regression, Random Forest, Multi-Layer Perceptrons) alongside creating a GRN-aware Graph Neural Networks (GNNs) using PyTorch Geometric — this demonstrated improved regression metrics (MSE, MAE, R^2) when biological prior knowledge was incorporated.

Advanced AI Mini-Project: TripAdvisor Restaurant Reviews Sentiment Analysis - Level 6

- Built a binary sentiment classification system achieving 93.3% accuracy (measured by F1-score and confusion matrix analysis) by implementing a custom Naive Bayes classifier from scratch with Laplace smoothing, preprocessing 4000 reviews using a NLP pipeline (tokenization, stemming, stop word removal).
- Implemented Logistic Regression, SVM, and Random Forest model's library to analyse and compare performance with Naïve Bayes, while exploring different ML models.
- Demonstrated strong algorithmic reasoning by selecting and justifying models based on performance trade-offs across F1-score, precision, and recall metrics.

Software Architecture Coursework – Healthcare Management System in Java– Level 6

- Architected a 3-tier healthcare application using MVC pattern, separating 14+ model classes and 4 controllers based on MVC pattern with separated Model, View, and Controller layers, and implementing the Singleton design pattern for centralized referral queue management to prevent duplicate processing.
- Built Java Swing GUI with full CRUD functionality across 5+ modules, by creating tabbed interfaces with JTable displays and form-based dialogs for real-time data operations.
- Implemented iterative Git-based development following Agile workflows and CSV-based data persistence.

UniMarketplace – Team Software Project – Level 4

- Led end-to-end development of a responsive eCommerce web application, managed the product lifecycle from requirements gathering and design through implementation, testing, and deployment. Utilised HTML, CSS, JavaScript, and Agile project management.
- Coordinated team discussions and delegated tasks to deliver a complex technical outcome as team lead.

HACKATHONS

HackLondon 2026 — Bush House, King's College London, Feb 2026 (22 hours)

- Built PropFlow, a retrofit intelligence platform addressing fragmented UK housing energy efficiency data, as part of a team of three.
- Led backend development, data integration, and scoring logic; delivered RetrofitIQ — a working tool that scores UK properties by retrofit priority using a composite score from open datasets including EPC data, HM Land Registry price-paid data, flood risk, and planning constraints.
- Integrated MHCLG EPC API, HM Land Registry, and Environment Agency flood data; handled real-world data quality challenges including inconsistent postcode formats, API inconsistencies, and frontend-backend integration issues.
- Tech stack: Python, Flask, MongoDB, React.js, Leaflet.js, Tailwind CSS.

Holistic AI x UCL Hackathon — Bloomsbury Campus, UCL, November 2025

- Built Stereotype Stompers, an AI-powered bias detection platform for healthcare text, as part of a team of three, competing in Track 2: Building Trustworthy Models for Stereotype Classification in Text Data.
 - Designed and implemented a multi-source data pipeline combining real scraped medical content, synthetic data generated via ChatGPT and Gemini AI, and the EMGSD dataset; fine-tuned GPT-2 on the combined corpus to produce realistic biased training data.
 - Developed a bias-detection model that flags stereotyped phrases in medical content (e.g. "Women exaggerate symptoms") and suggests fair, unbiased alternatives, with potential applications in healthcare AI auditing, medical education, and DEI compliance.
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TECHNICAL SKILLS

Language: Python, Java, JavaScript, HTML, CSS, SQL
Frameworks & Libraries: pandas, numpy, scipy, scikit-learn, matplotlib, PyTorch, NetworkX
Dev Tools: Git, GitHub, VS Code, Jupyter Notebook
Concepts: OOP, MVC, Design Patterns, REST APIs, Agile/Scrum, ML, Deep Learning, NLP, basic CI/CD.

CERTIFICATIONS & AWARDS

IBM Data Science Professional Certificate (In Progress - Coursera)	CSLA Level 2 Qualification – Community and Sports Leadership Award (Sept 2024)
Student Rep Award – Bronze and Silver Level , University of Hertfordshire Rep Excellence Program (2023–24)	Training for Adult Safeguarding in Physical Activity and Sport, by UK Coaching , Sept 2024
Go Herts Award – Bronze , University of Hertfordshire (Aug 2024)	FAA LEVEL 3 AWARD IN EMERGENCY FIRST AID AT WORK (RQF) By Nuco Training, Sept 2024
Introduction to Computer Science and Programming Specialization Certificate, Goldsmiths University of London (Jan 2023)	Certificate of Recognition by BCS: University of Hertfordshire Computer Science Chapter Member 23/24 supported by Hertfordshire Student Union.